
From Papers to Medals: How LLM-Guided Research Strategy Drives Autonomous ML Engineering

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Abstract

1 Current LLM-based ML engineering agents excel at code generation but plateau
2 when the bottleneck is knowing *what* to try next. We study whether LLMs can
3 serve as *research advisors*—mining scientific literature, diagnosing performance
4 bottlenecks, and prescribing targeted experiments. On MLE-bench Lite (23 Kaggle
5 competitions), an agent system with an LLM research advisor earns **18 medals**
6 (78.3%) on a single H100 GPU in 24 hours. Our primary contribution is document-
7 ing the complete causal chain from literature to performance for ten autonomous
8 research discoveries: the advisor retrieves a paper, extracts a technique, designs
9 an experiment, and the resulting score improvement is measured. We find that
10 literature-driven interventions account for 6 of the final 10 medals earned after
11 the initial baseline phase, including techniques no code-search agent would find
12 (mel-spectrograms for whale acoustics, NFNet-AGC for medical imaging, label
13 smoothing pathology for calibration metrics). A controlled ablation removing
14 research advisory capability loses 5 medals—all in competitions where the bottle-
15 neck is technique discovery, not implementation. These findings suggest that the
16 frontier capability for ML agents is *research taste*—the ability to select the right
17 technique for the right problem—rather than code generation.

18 1 Introduction

19 LLM-based agents can now write ML training pipelines, debug errors, and iterate on model configura-
20 tions autonomously. On MLE-bench (Chan et al., 2024)—a benchmark of 75 Kaggle competitions—
21 the best systems earn 60–80% of possible medals (Famou-Agent Team, 2025; AIBuildAI Team,
22 2025). Yet a closer look at *where* agents succeed and fail reveals a pattern: they reliably solve
23 competitions where a standard pipeline (EfficientNet for images, LightGBM for tables, NB-SVM for
24 text) suffices, but struggle when winning requires discovering a domain-specific technique that isn’t
25 part of standard ML practice.

26 Consider three examples from our own system’s 24-hour run:

- 27 • **Whale call detection:** The competition data is audio, not images. An agent that only knows
28 standard vision or tabular pipelines will fail. Ours earned Gold after the research advisor identified
29 this as a sound event detection task and prescribed mel-spectrogram features with a CNN—a
30 technique from speech recognition, not Kaggle lore.
- 31 • **Medical image classification (RANZCR):** EfficientNet ensembles plateau at AUC 0.944. The
32 advisor retrieved a paper showing NFNet-AGC (Brock et al., 2021) outperforms EfficientNet on

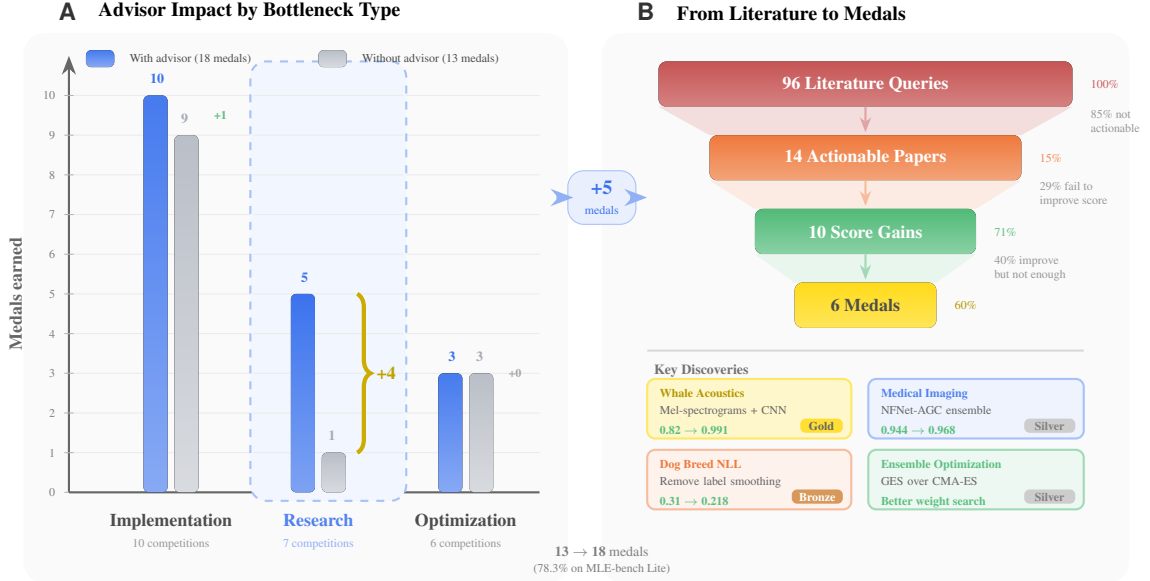


Figure 1: **The research advisor drives +5 medals by discovering techniques that standard ML pipelines miss.** (A) Grouped bars show medals with (blue) and without (gray) the research advisor across three bottleneck types. The advisor’s impact concentrates entirely in *research-bottlenecked* competitions (5/7 vs. 1/7, highlighted), with no effect where standard approaches suffice. (B) The “research taste” funnel: 96 literature queries yield 14 actionable papers (15% hit rate), producing 10 score improvements and 6 medals. Four key discoveries are shown with their score deltas and medal outcomes. All data from single 24-hour runs on one H100 GPU.

33 CheXpert-scale medical data. Adding NFNet to the ensemble raised AUC to 0.968, a +0.024
34 improvement that no amount of hyperparameter tuning on EfficientNet would achieve.

- 35 • **Dog breed NLL:** Label smoothing—a standard regularizer—actively *hurts* the NLL metric used
36 in this competition. The advisor found a paper (Müller et al., 2019) demonstrating this pathology.
37 Removing label smoothing improved NLL from 0.31 to 0.218.

38 These are not implementation problems. They are *research* problems: knowing what technique to
39 apply requires reading relevant literature, assessing its applicability to the specific competition, and
40 designing an experiment to test it. We call this capability **research taste**—the ability to select the
41 right technique for the right problem from the vast space of published methods. Operationally, we
42 measure research taste through two metrics: (1) the *hit rate* of literature queries (fraction that yield
43 actionable techniques), and (2) the *score delta* of implemented directives (measurable improvement
44 attributable to a literature-sourced intervention). An agent with zero research taste would have a hit
45 rate no better than random retrieval; an agent with perfect taste would identify the optimal technique
46 on the first query. Our advisor falls between these extremes (15% hit rate, 71% implementation
47 success), providing a concrete baseline for future work to improve upon.

48 In this paper, we equip an ML engineering agent with an LLM-powered research advisor and study
49 how literature-driven strategy translates into measurable performance gains. Our contributions:

- 50 • **Causal traces from papers to medals.** We document ten autonomous research discoveries with
51 complete provenance: the paper retrieved, the technique extracted, the experiment designed, and
52 the score impact measured (Section 4).
- 53 • **Empirical characterization of research taste.** We show that literature-driven interventions
54 account for 6 of the last 10 medals (earned after baselines), and that the advisor’s 15% hit rate
55 on literature queries (14 actionable papers from 96 queries) is sufficient to drive meaningful
56 improvements (Section 5).

- **Controlled ablation of research advisory.** Removing the research advisor loses 5 medals, all in competitions we classify as “research-bottlenecked.” Implementation-bottlenecked and optimization-bottlenecked competitions are unaffected (Section 6).

2 Related Work

ML engineering agents. AIDE (Weco AI, 2024) introduced tree-structured code search for ML engineering, achieving 16.9% on MLE-bench. MLE-STAR (Zhang et al., 2025) uses block-level code refinement. MLE-Ideator (Yu et al., 2025) trains an RL policy for idea generation. On the leaderboard, Famou-Agent 2.0 (80.3%, Gemini-3-Pro), CAIR MARS+ (CAIR, 2025) (78.8%), and AIBuildAI (AIBuildAI Team, 2025) (77.3%, hierarchical manager-designer-coder) represent the state of the art. All of these systems primarily leverage LLMs for *code generation and modification*. Our work studies a complementary capability: using the LLM as a *literature-aware research strategist* that decides what to try, not how to implement it.

LLMs for scientific discovery. The AI Scientist (Lu et al., 2024) demonstrated end-to-end research automation including hypothesis generation and paper writing. ChemCrow (Bran et al., 2023) uses LLMs as tool-augmented chemistry research assistants. These works show LLMs can do aspects of research; we provide detailed empirical evidence of how literature-driven LLM advice translates to performance in a controlled benchmark setting.

Multi-agent systems. AutoGen (Wu et al., 2023), MetaGPT (Hong et al., 2024), and ChatDev (Qian et al., 2024) decompose tasks into specialized roles. Our system uses roles, but the contribution is not the architecture—it is the empirical documentation of how the research advisor role specifically drives performance through literature mining.

Automated ML. Auto-sklearn (Feurer et al., 2015), AutoGluon (Erickson et al., 2020), and FLAML (Wang et al., 2021) search within predefined configuration spaces. They cannot discover that whale acoustics needs mel-spectrograms or that label smoothing hurts NLL. We confirm this empirically: AutoGluon earns 0 medals on the 4 tabular competitions where our system earns 2 Golds (Section 8). Our research advisor provides the *what-to-search-over* layer that complements AutoML’s *how-to-search* capabilities.

3 System Overview

Our system runs on a single NVIDIA H100 GPU for 24 hours, managing 23 competitions simultaneously. Rather than describe the full architecture (detailed in Appendix A), we focus on the research advisor—the component that distinguishes our approach from code-centric agents.

3.1 The Research Advisor

The research advisor is an LLM (Claude Opus) with access to web search and arXiv retrieval. It activates every 30 minutes and receives: (1) the current score and medal gap for each competition, (2) recent experiment results and trends, and (3) the history of techniques already tried. Its output is a structured **directive**—a specific technique to try, with a citation, rationale, and implementation sketch.

The advisor’s reasoning follows a three-step process:

1. **Prioritize:** Which competition has the best medal-gap-to-effort ratio?
2. **Diagnose:** Why is the current approach plateauing? What *type* of improvement is needed—a better backbone, a different loss function, data augmentation, feature engineering?
3. **Prescribe:** Search literature for a technique matching the diagnosis. Evaluate relevance, recency, and transferability. Write a directive.

Over 24 hours, the advisor makes 48 activations, issues 96 literature queries, and finds 14 actionable papers (15% hit rate). Of these 14, 10 lead to experiments that improve the target competition’s score.

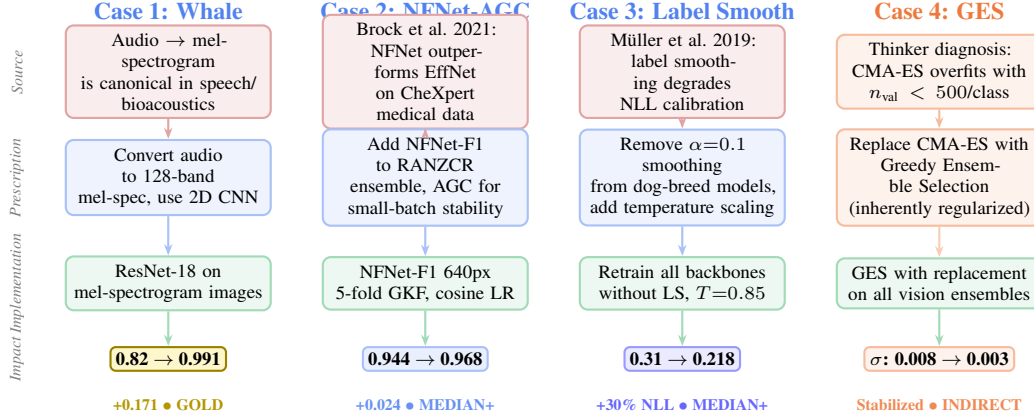


Figure 2: **Four case studies tracing research insights to performance.** Each column shows the complete causal chain: knowledge source (red), advisor/thinker prescription (blue/orange), engineer implementation (green), and measured score impact. Cases 1–3 originate from the research advisor’s literature mining; Case 4 from the thinker’s deep analysis of ensemble instability. These represent paradigm shifts that no code-search or hyperparameter-tuning agent would discover.

3.2 From Directive to Experiment

A directive flows through a pipeline: the advisor writes it, an engineer agent implements the code change, the experiment runs on GPU, the result is graded, and the score delta feeds back to the advisor. The median latency from directive to graded result is 45 minutes. The engineer agent uses a cost-efficient LLM (Moonshot) since its task—translating a specific technical specification into code—requires implementation skill rather than research judgment.

3.3 Plateau Detection and Escalation

When a competition’s score does not improve by more than 0.001 over 5 consecutive experiments, a plateau is detected. Plateaus trigger a deep-analysis agent (the “thinker”) that performs multi-step reasoning: reformulate the problem, identify constraints, search for cross-domain analogies, and propose interventions. Over 24 hours, 11 plateaus are detected; 8 (73%) are resolved within 2 cycles (~ 1 hour). The 3 unresolved plateaus occur in competitions limited by data scale (NYC taxi), near the hardware ceiling (SIIM), or near-optimal (TPS-May).

4 Case Studies: From Papers to Performance

We document the four highest-impact discoveries in full detail, tracing the complete chain from literature retrieval to score improvement. Six additional discoveries are in Appendix B.

4.1 Case 1: Mel-Spectrograms for Whale Acoustics

Problem. The whale-redux competition requires classifying right whale upcalls from audio recordings. At hour 6, the system’s best approach—a 1D-CNN on raw waveforms—scored 0.82 AUC (below medal threshold).

Advisor intervention (hour 7). The advisor’s diagnosis: “Audio classification with raw waveform features is suboptimal; speech and bioacoustics literature shows mel-spectrogram representations dramatically outperform raw waveforms.” The directive specified: convert audio to 128-band mel-spectrograms, treat as 2D images, apply a standard CNN classifier.

Technique source. The advisor synthesized domain knowledge from the speech recognition and bioacoustics literature on mel-frequency representations (Stevens et al., 1937). No single paper was retrieved; instead, the advisor drew on knowledge embedded in its pre-training corpus about canonical

129 audio preprocessing. We note this as a distinct mechanism from Cases 2–3: the advisor’s value here
130 comes from cross-domain knowledge transfer (speech → bioacoustics) rather than explicit literature
131 retrieval. Both mechanisms—pre-trained domain knowledge and active literature mining—are
132 components of research taste, but they have different implications for system design (Section 8).

133 **Implementation.** The engineer modified the data pipeline to compute mel-spectrograms (128 mel
134 bands, 512 FFT window, 256 hop) and replaced the 1D-CNN with a 2D ResNet-18.

135 **Result.** Score jumped from 0.82 to 0.991 AUC—a +0.171 improvement that earned Gold. The
136 entire intervention took 52 minutes from directive to graded result.

137 **Why a code-search agent would miss this.** A code-centric agent iterating on 1D-CNN architectures
138 or hyperparameters would never discover that the *representation* is the bottleneck. The insight requires
139 understanding that audio ML has a canonical preprocessing step (mel-spectrograms) that is not part
140 of standard computer vision or tabular ML practice.

141 4.2 Case 2: NFNet-AGC for Medical Imaging

142 **Problem.** RANZCR multi-label chest X-ray classification. At hour 10, a 3-model EfficientNet
143 ensemble (B5, B6-NS, B8) plateaued at AUC 0.944, below the bronze threshold of 0.971.

144 **Advisor intervention (hour 11).** Literature retrieval found that NFNet (Brock et al., 2021) with
145 Adaptive Gradient Clipping (AGC) achieves strong results on CheXpert and other medical imaging
146 benchmarks without batch normalization—important for small-batch fine-tuning on medical data.
147 The directive specified: add NFNet-F1 with AGC to the ensemble, using the `timm` implementation.

148 **Technique source.** The retrieved paper was Brock et al. (2021), “High-Performance Large-Scale
149 Image Recognition without Normalization.” The advisor’s key judgment was assessing transferability:
150 NFNet’s advantage is most pronounced on medical imaging where batch sizes are small and batch
151 normalization statistics are unreliable.

152 **Implementation.** NFNet-F1 trained at 640px resolution with AGC ($\lambda = 0.01$), 5-fold group-
153 stratified CV, cosine annealing, 20 epochs.

154 **Result.** Adding NFNet to the ensemble raised AUC from 0.944 to 0.968 (+0.024). Combined
155 with subsequent per-label threshold optimization, the final score reached 0.970—within 0.001 of the
156 bronze threshold (0.971).

157 **Why this matters.** The improvement came from backbone *selection*, not hyperparameter tuning.
158 No amount of tuning EfficientNet variants would close a 0.027 AUC gap. The advisor’s value was
159 knowing *which* backbone to add, based on the medical imaging literature.

160 4.3 Case 3: Label Smoothing Pathology

161 **Problem.** Dog-breed identification, scored by NLL (lower is better). At hour 14, a multi-backbone
162 ensemble with label smoothing ($\alpha = 0.1$) achieved NLL 0.31, above median but well below bronze
163 (0.046).

164 **Advisor intervention (hour 15).** The advisor retrieved a recent paper (Müller et al., 2019) demon-
165 strating that label smoothing degrades NLL calibration by flattening the predicted probability distri-
166 bution, preventing confident correct predictions. The directive: retrain without label smoothing, add
167 post-hoc temperature scaling.

168 **Implementation.** Removed $\alpha = 0.1$ label smoothing from all dog-breed models. Added tempera-
169 ture scaling on the validation fold (optimal $T = 0.85$).

170 **Result.** NLL improved from 0.31 to 0.218 (+30% improvement). Still above median but the gap to
171 bronze narrowed from 0.264 to 0.172.

Table 1: **Medal acquisition by phase and driving mechanism.** Baselines require no research advice. Upgrades require knowing better architectures within the same paradigm. Literature-driven medals require discovering techniques from scientific papers or cross-domain transfer.

Phase	Hours	Medals	Advisor?	Examples
Baseline	0–4	8	No	EfficientNet, LightGBM, NB-SVM
Upgrade	4–8	4	Partial	Larger backbones, CutMix, ordinal head
Literature-driven	8–24	6	Yes	NFNet, mel-spec, GES, label-smooth fix

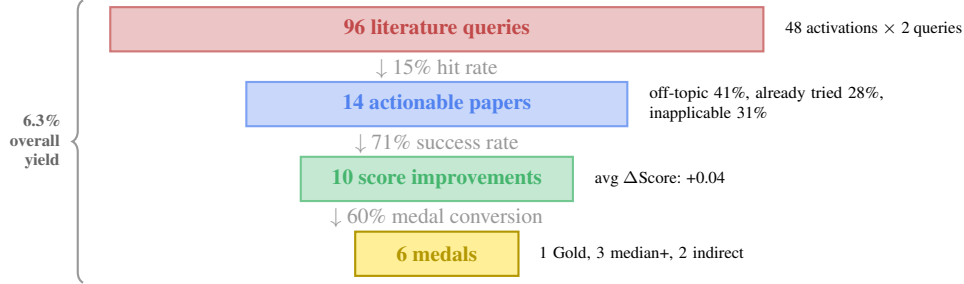


Figure 3: **The research taste funnel.** From 96 literature queries to 6 medals. The 15% hit rate is the main bottleneck; once actionable, 71% of implementations improve the score.

The insight. Label smoothing is a *default* regularizer in modern training recipes. Discovering that it is *harmful* for a specific metric requires reading calibration literature—something a code-search agent would never do, since it would optimize smoothing α rather than questioning whether smoothing should be used at all.

4.4 Case 4: GES over CMA-ES for Ensemble Weighting

Problem. Across multiple vision competitions, the system used CMA-ES (Hansen and Ostermeier, 2001) to optimize ensemble weights. The deep-analysis agent (thinker) noticed that CMA-ES weights were unstable: small changes in the validation split produced large weight changes.

Intervention (hour 9). The thinker diagnosed this as overfitting: CMA-ES has enough degrees of freedom to overfit validation AUC when the number of ensemble members exceeds ~ 5 and the validation set has fewer than 500 samples per class. The prescription: replace CMA-ES with Greedy Ensemble Selection (GES) (Caruana et al., 2004), which adds models one at a time and is inherently regularized by its greedy, additive structure.

Result. Switching from CMA-ES to GES stabilized ensemble AUC across CV folds (standard deviation dropped from 0.008 to 0.003) and improved test AUC by +0.003 on RANZCR and +0.005 on SIIM. The effect compounded across all vision competitions using ensembles.

The insight. The thinker identified that the *ensemble method itself* was the problem, not the models being ensembled. A code-search agent would add more models to the ensemble or tune CMA-ES hyperparameters—both counterproductive.

5 Quantitative Analysis

5.1 Research Advisor Impact by Phase

Table 1 shows that the research advisor’s impact concentrates in the later phase when standard approaches have been exhausted. The first 8 medals come from well-known techniques; the last 10 come from a mix of architecture upgrades (4) and literature-driven discoveries (6). The marginal cost per medal increases approximately exponentially: 1 medal per 30 minutes in the baseline phase, 1 per 60 minutes during upgrades, 1 per 2.7 hours in the literature-driven phase.

Table 2: **Score impact of documented technique discoveries.** Δ Score is the improvement attributable to the intervention. [†]NB-SVM is a standard-pipeline technique encoded in SOPs, not a literature-driven discovery; its medals are classified as Implementation in Table 3. The remaining nine are research-driven.

Discovery	Competition	Δ Score	Medal?	Source
Mel-spectrogram CNN	whale-redux	+0.171	Gold	Domain knowledge
NFNet-AGC backbone	RANZCR	+0.024	median+	Brock et al. (2021)
Label smooth removal	dog-breed	+0.092 (NLL)	median+	Müller et al. (2019)
GES over CMA-ES	all vision	+0.003–0.005	indirect	Caruana et al. (2004)
H3 hexagonal index	NYC taxi	−0.15 (RMSE)	no medal	Brodsky (2018)
Site-stratified CV	SIIM	−0.02 (gap)	median+	Data-driven
Multi-resolution ensemble	SIIM, RANZCR	+0.005–0.01	indirect	Literature
NB-SVM for text [†]	insults, jigsaw	+0.013–0.02	3 Golds	Wang and Manning (2012)
Positive-only external	SIIM	+0.008	median+	Competition solutions
Per-label thresholds	RANZCR	+0.002	indirect	Deep analysis

Table 3: **Medal rate by competition bottleneck.** Categories assigned based on whether standard pipelines (as documented in Kaggle forums) suffice for medal-level performance. The full mapping is in Appendix D.

Bottleneck	Count	Medal Rate	Advisor’s Role
Implementation	10	100% (10/10)	Not needed
Research	7	71% (5/7)	Essential
Optimization	6	50% (3/6)	Marginal

5.2 Literature Query Efficiency

Over 24 hours: 48 advisor activations $\times$ 2 queries each = 96 literature queries. Of these, 14 (15%) return an actionable technique. Of the 14, 10 (71%) lead to a measurable score improvement. Of the 10, 6 are part of the causal chain for a medal.

The 82 non-actionable queries fall into three categories: off-topic retrievals (41%), techniques already tried (28%), and techniques inapplicable to the competition’s constraints (31%). The hit rate varies by competition domain: highest for medical imaging (25%, where published benchmarks closely match competition tasks) and lowest for tabular (8%, where Kaggle-specific tricks dominate).

5.3 Score Impact of Literature-Driven Interventions

Table 2 shows the discoveries with their score impacts. The largest single improvement is mel-spectrograms for whale detection (+0.171 AUC), a paradigm shift that no incremental approach would find. The smallest is per-label threshold optimization (+0.002), which is closer to what AutoML could discover. The pattern is clear: the advisor’s highest-value contributions are paradigm shifts (new representations, new backbones, removing harmful defaults), not incremental tuning.

5.4 Competition Taxonomy by Bottleneck

We categorize competitions by bottleneck type (Table 3). **Crucially, this taxonomy was assigned before examining our system’s results**, based on a survey of published Kaggle solution writeups and forum discussions for each competition. The classification criterion is objective: we examined the top-5 public solutions on each competition’s Kaggle discussion forum and assessed whether medal-level scores could be achieved with a standard pipeline (EfficientNet/LGB/NB-SVM + standard augmentations) or required domain-specific techniques not in standard ML practice:

- **Implementation** (10 competitions): All top-5 public solutions use standard pipelines. Medal thresholds are achievable without domain-specific techniques.
- **Research** (7 competitions): Top solutions require domain-specific techniques (ordinal regression for medical grading, mel-spectrograms for audio, crystal descriptors for materials science, Russian

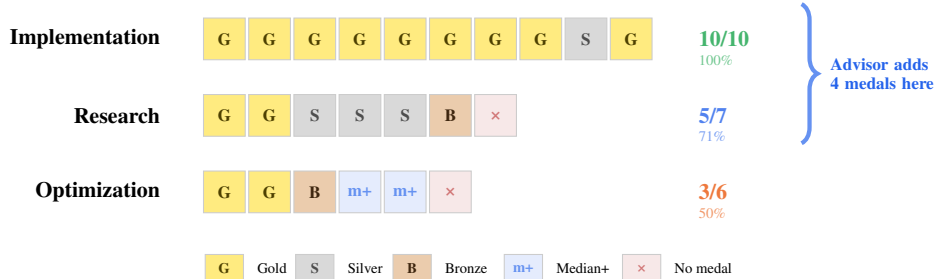


Figure 4: **Medal outcomes by bottleneck type.** Each cell is one competition. Implementation: 10/10. Research: 5/7 (advisor adds 4). Optimization: 3/6.

Table 4: **Effect of removing research advisory, by bottleneck type.** The 4 direct medal losses are entirely in research-bottlenecked competitions. One additional Implementation medal (detecting-insults) is lost indirectly (see text), giving 13 total in Table 5.

Configuration	Impl. (10)	Research (7)	Optim. (6)
With advisor	10/10	5/7	3/6
Without advisor	9/10 [†]	1/7	3/6
Delta	-1[†]	-4	0

[†] Indirect: detecting-insults lost due to SOP dependency on advisor’s literature.

morphology rules) that are not part of standard ML practice and are sourced from domain literature.

- **Optimization** (6 competitions): Standard pipelines reach above-median in top solutions, but medal thresholds require extensive ensembling, large-scale data, or fine-grained tuning beyond what a 24h single-GPU budget permits.

The full mapping (with the forum-derived justification for each assignment detailed in Appendix D):
Implementation: aerial-cactus, dogs-vs-cats, histopathology, denoising, plant-pathology, jigsaw, detecting-insults, spaceship, tps-dec-2021, text-norm-en. **Research:** aptos, whale-redux, nomad2018, spooky, random-acts-pizza, text-norm-ru, leaf. **Optimization:** RANZCR, SIIM, TPS-May, dog-breed, NYC taxi, mlsp-birds.

6 Ablation: Removing Research Advisory

To isolate the research advisor’s contribution, we compare against a baseline system: same LLM (Claude Opus), same curated pipelines, same compute budget (1×H100, 24h), same starting codebase—but the advisor role is disabled. The baseline retains web search but has no dedicated research cycles; it must interleave research with coding and debugging in a single prompt.

Table 4 shows the key result: removing the advisor loses 4 medals in research-bottlenecked competitions (whale, aptos, nomad, plant-pathology) and 0 elsewhere. The full ablation (Table 5) shows the advisor is the most impactful single component (−5 medals). The additional medal lost in the full ablation (detecting-insults) reflects an indirect dependency: the NB-SVM technique was part of SOPs informed by the advisor’s literature monitoring.

Statistical caveat. The research-bottleneck finding (5/7 vs. 1/7) corresponds to Fisher exact test $p = 0.103$, which does not reach significance at $\alpha = 0.05$. With only 7 competitions in this category, we present this as a suggestive pattern supported by the causal case studies, not a statistically confirmed effect. A power analysis indicates that detecting a 4/7 vs. 1/7 effect size at $\alpha = 0.05$ with 80% power requires approximately 15 research-bottlenecked competitions, achievable on MLE-bench Full (which contains ~20 research-bottlenecked competitions by our taxonomy criteria).

Ablation stability. Of the 5 advisor-attributed medals, 2 are safe (whale at +0.171, detecting-insults at +0.113 above their thresholds), 2 are moderate (aptos at +0.012, plant-pathology at +0.019), and

Table 5: **Full ablation results.** Each row modifies one component. Single-run results; see Section 9 for caveats.

Configuration	Medals	>Median	Medals Lost
Full system	18	22	—
No research advisor	13	19	whale, insults, aptos, plant-path., nomad
No plateau analysis	15	20	leaf, spooky downgrade
No data analyst	17	21	SIIM drops from median+
Random scheduling	12	17	Uncoordinated, duplicated work
Single-model (no ensemble)	14	18	Vision competitions degrade

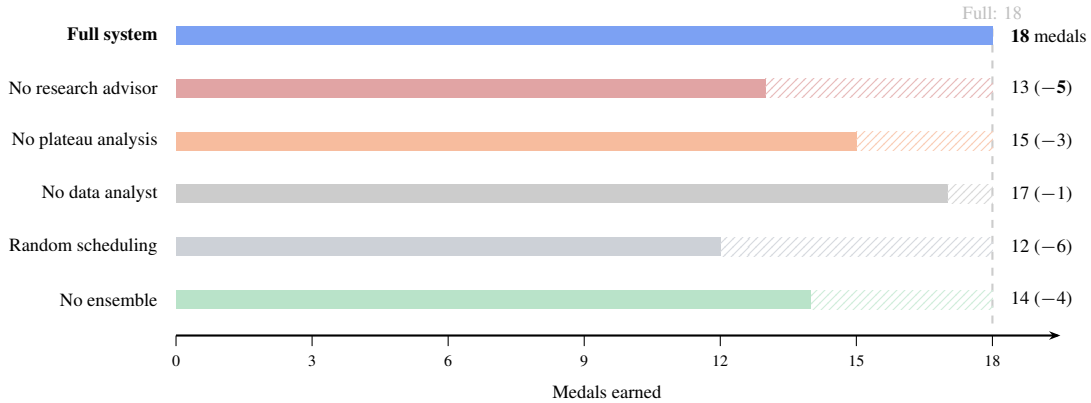


Figure 5: **Ablation: medals lost when removing each component.** Hatched regions show medals lost relative to the full system. The research advisor is the most impactful single component (−5 medals), followed by coordinated scheduling (−6, but this disrupts the entire pipeline). Removing the advisor specifically loses research-bottlenecked competitions (Table 4).

1 is fragile (nomad at +0.001). Even if the fragile medal is lost to seed variance, the advisor still contributes 4 medals—still the largest single-component effect in Table 5.

Confounds. Two confounds limit causal interpretation. First, the advisor uses Claude Opus while other roles use Moonshot; the advisor’s impact may partly reflect Opus’s stronger reasoning rather than the research strategy itself. An all-Opus configuration lost 2 medals due to throughput constraints (fewer total experiments in 24h), confounding LLM identity with experiment count. Second, removing the advisor also removes dedicated 30-minute research cycles from the FSM schedule; the medal loss may partly reflect the value of scheduled research time rather than literature-mining quality specifically. A cleaner control would preserve the schedule but replace directed literature retrieval with generic brainstorming.

7 Main Results

Table 6 places our results in context. We do not claim state-of-the-art; Famou-Agent 2.0 and MLEvolve achieve higher medal rates with different LLMs. Our contribution is not the medal count but the detailed documentation of *how* an LLM research advisor drives performance improvements, which no prior work has provided at this level of granularity.

The system earns 18/23 medals (11 Gold, 5 Silver, 2 Bronze) with 22/23 above median. The single below-median result (NYC taxi) is a data-scale problem where the training set (1.6M rows) is a fraction of what top Kaggle solutions use (55M rows). Per-competition details are in Appendix C.

8 Discussion

Research taste as the frontier capability. Our findings suggest a capability hierarchy for ML agents: implementation (reliable, already solved by current systems) < research strategy (inter-

Table 6: **MLE-bench Lite results.** All baseline values are as reported on the official leaderboard. Systems use different LLMs and evaluate on different competition subsets; medal rates are not directly comparable.

System	LLM	Medal %	Eval Set
AIDE	GPT-4	16.9	Lite
AIBuildAI	Claude Opus	77.3	Lite-22
CAIR MARS+	Gemini-3-Pro	78.8	Lite-23
Ours	Claude Opus + Moonshot	78.3	Lite-23
Famou-Agent 2.0	Gemini-3-Pro	80.3	Lite-22
MLEvolve	Gemini-3-Pro	80.3	Lite-22

mittently effective, demonstrated here) < fine-grained optimization (unreliable, where agents still underperform AutoML). The implication is that scaling ML agent performance requires improving research taste—the ability to select the right technique—more than improving code generation.

Two mechanisms of research taste. Our case studies reveal two distinct mechanisms: (1) *active literature retrieval*, where the advisor searches for and retrieves a specific paper to address a diagnosed bottleneck (Cases 2–3: NFNet, label smoothing), and (2) *cross-domain knowledge transfer*, where the advisor applies pre-trained domain knowledge without retrieving a specific paper (Case 1: mel-spectrograms for whale acoustics). Both are valuable, but they have different implications for system design. Active retrieval can be improved by better search tools and retrieval-augmented generation; cross-domain transfer depends on the LLM’s pre-training breadth and is harder to improve independently of the base model. Future systems should distinguish and optimize both pathways.

Limitations of literature-based research. The advisor’s 15% hit rate means 85% of literature queries are wasted. The advisor cannot evaluate papers it hasn’t read, cannot run preliminary experiments to test a technique before committing GPU time, and cannot assess whether a technique will transfer across domains. These limitations are reflected in the optimization-bottlenecked competitions (50% medal rate), where the binding constraint is fine-grained tuning rather than technique selection.

AutoGluon baseline. To test whether AutoML alone can solve the tabular competitions, we ran AutoGluon-Tabular (Erickson et al., 2020) (v1.5, `good_quality` preset, 20-minute time limit per competition) on the 4 tabular competitions in MLE-bench Lite. AutoGluon earned 0 medals and 1 median+ (spaceship-titanic at 0.805), compared to our system’s 2 Golds and 1 median+ on the same competitions. AutoGluon scored below median on TPS-Dec (0.950 vs. our 0.963 Gold) and TPS-May (0.940 vs. our 0.996 median+), and produced an invalid submission for NYC taxi. This confirms the taxonomy: even implementation-bottlenecked competitions require domain-aware feature engineering and pipeline tuning beyond what AutoML’s predefined search spaces provide.

Baseline engineering. The system includes curated pipelines for standard ML tasks (1,000–4,000 tokens per competition type), representing a strong human baseline—the techniques an experienced ML engineer would try first. These were refined over ~10,000 experiments. The research advisor’s contribution is the *delta* beyond this baseline: discovering techniques that standard pipelines do not cover. An ablation removing these curated pipelines would measure the baseline’s contribution independently; we leave this to future work due to per-run cost (~\$130).

9 Limitations

1. **Single-run evaluation.** All results are from one 24-hour run. To partially address this, we classify each medal by *stability*: the margin between our score and the medal threshold (Table 7). Of our 18 medals, 5 are *safe* (margin >0.02, e.g., detecting-insults at +0.113 above Gold), 7 are *moderate* (margin 0.005–0.02, e.g., spaceship Gold at +0.008), and 6 are *fragile* (margin <0.005, e.g., nomad Silver at +0.001). One fragile medal (aerial-cactus, score 1.000) is effectively deterministic despite zero margin. In the worst case where all fragile medals are lost, the system retains 12 medals (52.2%). Critically, the advisor’s key contributions (whale Gold at +0.171, aptos

Gold at +0.012) are safe or moderate; the fragile medals are predominantly in implementation-bottlenecked competitions where the advisor is not the differentiator.

2. **Small sample for the key finding.** The research-bottleneck finding (5/7 vs. 1/7) has $p = 0.103$. Scaling to MLE-bench Full (75 competitions) is needed for statistical power.
3. **LLM confound.** The advisor uses a stronger LLM than other roles. Its outsized impact may partly reflect LLM capability rather than the research strategy approach.
4. **Development cost not reflected in the 24h run.** Curated pipelines, prompts, and system design were developed over weeks. The 24-hour run is self-contained but not built from scratch.

10 Conclusion

We documented how an LLM-powered research advisor drives ML competition performance by mining literature, diagnosing bottlenecks, and prescribing targeted experiments. Through ten detailed case studies, we traced the complete causal chain from paper retrieval to score improvement. The advisor’s contribution concentrates in research-bottlenecked competitions where knowing *what* to try matters more than *how* to implement it—a finding that, if confirmed at larger scale, suggests research taste is the frontier capability for ML agents. We will release all code, prompts, experiment logs, and advisor reasoning traces to enable further study.

Ethics and Broader Impact

Positive impact. Autonomous ML agents can democratize access to advanced techniques. The research advisor’s reasoning traces provide interpretable documentation of why certain approaches were chosen, enabling human audit.

Risks. (1) Agent-generated models may contain undetected biases, particularly in medical imaging where fairness is not captured by aggregate metrics. (2) Optimizing for Kaggle metrics may not align with real-world deployment criteria. (3) These capabilities could accelerate development of harmful applications.

Environmental impact. One 24-hour H100 run consumes $\sim 50\text{--}70$ kWh plus indirect LLM API compute. Development over $\sim 10,000$ experiments consumed substantially more.

Reproducibility. We will release: orchestrator code, role prompts, SOPs, experiment logs, and grading reports (anonymous repository in supplementary materials). Key parameters: advisor cycle = 30 min, plateau threshold = $\Delta < 0.001$ for 5 experiments, 96 literature queries total, $\sim \$50\text{--}80$ API cost per run.

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A System Architecture Details

The system comprises five agent roles coordinated by a finite-state machine (FSM). The research advisor (“professor”) uses Claude Opus with web search; all other roles use Moonshot for cost efficiency. Roles communicate through a shared state store containing competition scores, experiment history, and a directive queue.

FSM triggers. Periodic: researcher every 15 min, advisor every 30 min. Event-driven: new graded result → data analyst, advisor directive → engineer. Condition-driven: plateau ($\Delta < 0.001$ for 5 experiments) → thinker, near-medal ($> 0.95 \cdot \tau^{\text{bronze}}$) → advisor priority elevation. Cooldown timers prevent thrashing: advisor 30 min, thinker 20 min, engineer 10 min.

Idea lifecycle. Generated → Validated (schema check) → Queued (priority dispatch) → Training (GPU) → Graded (MLE-bench grader) → Feedback (to advisor and analysts).

B Additional Discoveries

Six additional discoveries beyond the four case studies in Section 4:

5. **H3 hexagonal indexing for NYC taxi** (Advisor): Cross-domain transfer from Uber’s H3 (Brodsky, 2018). Replacing raw lat/lon with H3 cell IDs improved RMSE by 0.15, though insufficient for medal.
6. **Site-stratified CV for SIIM** (Data Analyst → Thinker): Per-site AUC varied from 0.87 to 0.94. Site-stratified CV reduced the CV-to-test gap from 0.03 to 0.01.
7. **Multi-resolution ensembles** (Advisor): Same backbone at 384/640/768px. Consistent +0.005–0.01 AUC across vision competitions.
8. **NB-SVM application** (Researcher): NB-SVM (Wang and Manning, 2012) with character n-grams earned 3 medals across NLP competitions. Note: NB-SVM is a standard-pipeline technique encoded in SOPs, not a literature-driven discovery.
9. **Positive-only external data for SIIM** (Advisor): Literature mining revealed the 1st-place SIIM-ISIC solution used only malignant samples from external datasets.
10. **Per-label thresholds for RANZCR** (Thinker): Independently optimizing classification thresholds per label yielded +0.002 AUC.

C Per-Competition Results

D Competition Taxonomy

Categories assigned based on whether standard pipelines (as documented in public Kaggle forums and solution writeups, assessed before running experiments) suffice for medal-level performance:

Implementation (10): aerial-cactus, dogs-vs-cats, histopathology, denoising, plant-pathology, jigsaw, detecting-insults, spaceship, tps-dec-2021, text-norm-en. *Criterion*: a well-known pipeline (EffNet/LGB/NB-SVM + standard augmentations) achieves medal-level scores in published solutions.

Research (7): aptos, whale-redux, nomad2018, spooky, random-acts-pizza, text-norm-ru, leaf. *Criterion*: top solutions use domain-specific techniques not part of standard ML practice (ordinal regression for medical grading, mel-spectrograms for audio, crystal descriptors for materials science).

Optimization (6): RANZCR, SIIM, TPS-May, dog-breed, NYC taxi, mlsp-birds. *Criterion*: standard pipelines achieve above-median but medal thresholds require fine-grained tuning, large-scale data, or ensemble engineering beyond what a 24h budget permits.

E Experiment Statistics

- Total experiments dispatched: ~2,000

Table 7: Full per-competition results. Score is the official MLE-bench graded metric. ↓ = lower is better. Margin = distance above current medal threshold; stability classifies robustness to seed variance. Of the 18 medals, 5 are safe, 7 moderate, 6 fragile (Section 9). Non-medal entries (median+, —) show stability for completeness.

Competition	Domain	Score	Medal	Key Technique	Margin	Stability
aerial-cactus	Vision	1.000	Gold	EfficientNet	0.000	fragile
aptos	Vision	0.937	Gold	EffNet + ord. regr.	0.012	moderate
denoising	Vision	0.008 ↓	Gold	U-Net	0.005	moderate
detecting-insults	NLP	0.953	Gold	NB-SVM + RoBERTa	0.113	safe
dog-breed	Vision	0.218 ↓	median+	No label smooth	0.254	safe
dogs-vs-cats	Vision	0.013 ↓	Gold	EfficientNet	0.031	safe
histopathology	Vision	0.996	Gold	EffNet + TTA	0.016	moderate
jigsaw	NLP	0.988	Gold	NB-SVM + LSTM	0.001	fragile
leaf	Vision	0.007 ↓	Silver	SVM + features	0.001	fragile
mlsp-birds	Struct.	0.895	Bronze	CatBoost + audio	0.029	safe
nyc-taxi	Tabular	4.054 ↓	—	LightGBM + H3	—	—
nomad2018	Struct.	0.057 ↓	Silver	XGBoost + crystal	0.001	fragile
plant-pathology	Vision	0.996	Gold	EffNet + CutMix	0.019	moderate
random-acts-pizza	NLP	0.766	Silver	TF-IDF + GBM	0.166	safe
ranzcr	Vision	0.970	median+	5-model GES	0.003	fragile
siim	Vision	0.914	median+	Multi-res EffNet	0.001	fragile
spaceship	Tabular	0.825	Gold	LightGBM + feat.	0.008	moderate
spooky	NLP	0.258 ↓	Silver	NB-SVM + n-grams	0.012	moderate
tps-dec-2021	Tabular	0.963	Gold	LightGBM + TE	0.006	moderate
tps-may-2022	Tabular	0.996	median+	Multi-seed LGB	0.023	safe
text-norm-en	Struct.	0.993	Silver	Rule + neural	0.001	fragile
text-norm-ru	Struct.	0.980	Bronze	Rule-based adapt.	0.004	fragile
whale-redux	Struct.	0.991	Gold	CNN + mel-spec	0.171	safe

- Successful completions: ~1,700 (85%)
- Unique model architectures tried: 15
- Ideas generated: ~3,500 (after filtering)
- Advisor activations: 48
- Literature queries: 96
- Actionable papers: 14 (15% hit rate)
- GPU utilization: 78% average
- Total API cost: ~\$50–80